Soumodeep Karmakar

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Education

B.E. Electrical Engineering 2024 – 2028 (Expected)
Jadavpur University, Kolkata | CGPA: 7.5 (upto 4th sem) *Kolkata, India*

Skills

Languages & Core CS: Python, C/C++, JavaScript, TypeScript, SQL, REST APIs, HTTP, OOP, Data Structures & Algorithms
Backend & Full-Stack: FastAPI, React, PostgreSQL, SQLite (WAL), Docker, Linux, Git, Concurrency (asyncio), pytest, AWS (EC2)
AI & Systems: Deep Learning, PyTorch, Mixture-of-Experts, LLMs, Transformer Architectures, Context Compaction, RAG

Projects

Durable Tool-Execution Runtime | *Python, asyncio, SQLite, POSIX* |

- Engineered a high-throughput execution backend for tool calls with single-writer **SQLite WAL** persistence, persistent commit-before-ack boundaries, and **argument-hashed idempotency** to guarantee zero duplicate execution.
- Implemented **epoch-fenced worker leases** with heartbeat renewal and crash recovery; validated via **560 automated tests**, process-kill (**SIGKILL**) chaos testing, and sustained **2,115 jobs/s** throughput (**3.63 ms** p99 admission latency).

AI Customer Support | *FastAPI, React, TypeScript, PostgreSQL, Docker* |

- Built a full-stack asynchronous automation service using **FastAPI**, **PostgreSQL**, and Docker to ingest, triage, and route high-volume email event streams with duplicate-reply protection and connection pooling (`psycopg_pool`).
- Implemented PostgreSQL full-text search with deterministic business rules and shipped a **React + TypeScript** admin dashboard for human-in-the-loop resolution, verified by **39 offline unit and API tests**.

ReAct Agent Framework | *Python, FastAPI, Modal, SWE-bench* |

- Built a modular **ReAct agent framework** with a code-debugging agent (sandboxed shell execution, git patch export) and a chess agent running programmatic lookahead search inside an isolated container.
- Designed a **working-memory context compaction** engine; resolved SWE-bench Django task django-15368 while cutting peak prompt size by **86.2%** (35.7K → 4.9K tokens) and total tokens by **86.3%**.

nanoMoE | 120M params (65.8M active) | *PyTorch, MoE, MLA, Modal* |

- Built a **sparse Mixture-of-Experts LLM from scratch** with **DeepSeek-style Multi-head Latent Attention (MLA)** for KV-cache compression, RoPE, and SwiGLU.
- Designed a **Top-2 noisy-gating router** with auxiliary load-balancing loss (four experts per layer); pre-trained on **FineWeb-Edu** on cloud H100 GPUs via Modal.

Experience

Research Contributor Jun 2026 – Present
CMATER Lab, Jadavpur University (under Prof. Ram Sarkar) *Kolkata, India*

- Built a cross-dataset image segmentation pipeline using **SAM, CLIP, and CoOp** to generalize across unseen imaging domains.
- Developed **CLIP-guided mask selection** and pseudo-labeling for domain shift, contributing to **two papers** (one accepted at *CVIP 2026*, one first-author under review at *Knowledge-Based Systems*).

Publications

- **Calibrated SAM-CLIP Pseudo-Labeling and Cross-Dataset CutMix for Few-Shot Medical Image Segmentation.** **S. Karmakar**, S. Pani, S. Ghosh, J. Banerjee, R. Pal, R. Sarkar. *Knowledge-Based Systems (Elsevier).* **Submitted (under review)**.
- **Beyond Spatial Prototypes: Joint Spatial–Spectral Prototype Learning for Few-Shot Medical Image Segmentation.** J. Banerjee, R. Pal, S. Ghosh, S. Pani, **S. Karmakar**, R. Sarkar. *CVIP 2026.* **Accepted**.

Achievements

AISEHack 2026 – Top 4 Team, IIIT Hyderabad (IBM Theme: Flood Detection) Apr 2026
Medha-Manthan 2025 – 1st Prize, IEEE Kolkata Chapter (assistive communication board) Sep 2025
Hackforge – 3rd Position, Srijan'25, CodeClub Jadavpur University Mar 2025